

Initial and final

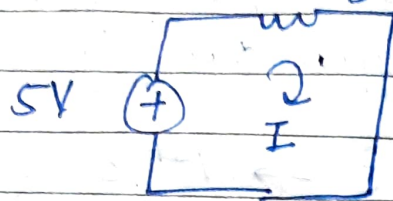
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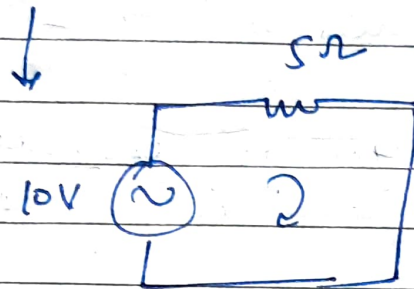
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Condition and procedure
of Evaluation

Transient Response:-

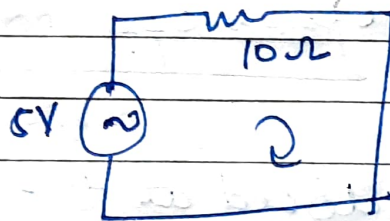


$$I = \frac{5V}{5\Omega} = \frac{V}{R} = 1 \text{ Amp}$$



$$I = \frac{10}{5} = 2 \text{ Amp}$$

or



$$I = \frac{5}{10} = 0.5 \text{ Amp.}$$

The transition period during which the branch current and element voltage change from their former values to new ones. This period is called the Transient. After the transient has passed, the ckt is said to be in steady state.

The linear differential equation that describes the circuit will have 2 parts of solution:-

- ① C.F. → corresponding to transient state
- ② P.I. → " " steady state

A circuit containing L or C and R will have current and voltage variables given by differential eqn. It is linear 1st order differential eqn.

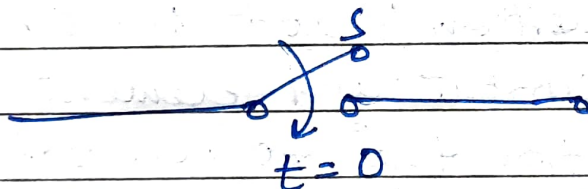
R-L, R-C $\rightarrow$ 1st order ckt

R-L-C, series or || $\rightarrow$ 2nd order ckt.

The ckt. changes are assumed to occur at time $t = 0$ and is represented by switch

Switch close $\rightarrow$ on } at $t = 0$
Switch open $\rightarrow$ off }

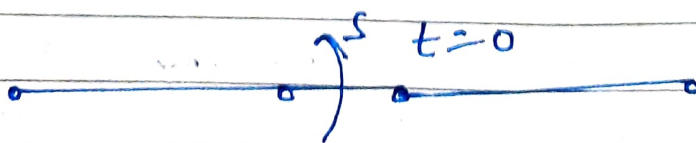
(a) Switch is closed at $t = 0$



$t = 0^-$, the instant prior to $t = 0$

$t = 0^+$, the " immediately after switching

(b) Switch is opened at $t = 0$



Differential Eqn :-

① 1st order Homogeneous differential Eqn :-

$$\frac{dy(t)}{dt} + Py(t) = 0$$

soln :- $y(t) = Ke^{-Pt}$

② 1st order non-homogeneous differential Eqn

$$\frac{dy(t)}{dt} + Py(t) = Q$$

soln :- $y(t) = \frac{Q}{P} + Ke^{-Pt}$

③ 2nd order differential Eqn :-

$$A \frac{d^2y(t)}{dt^2} + B \frac{dy(t)}{dt} + Cy(t) = 0$$

soln :- $y(t) = K_1 e^{p_1 t} + K_2 e^{p_2 t}$

If $p_1 = p_2$,

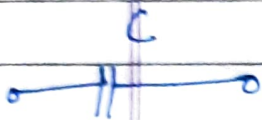
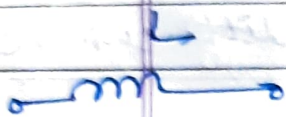
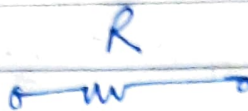
then
soln :- $y(t) = K_1 e^{p_1 t} + K_2 t e^{p_1 t}$

Initial condition in circuits :-

Element with Initial Condⁿ

Eq. ckt at $t=0^+$

Eq. ckt at $t=\infty$



Various Responses :-

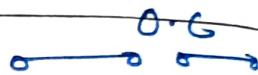
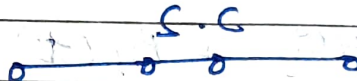
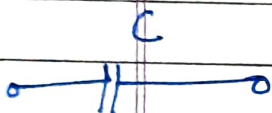
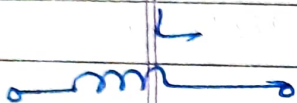
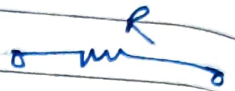
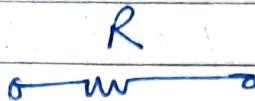
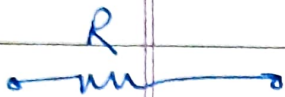
- ① Transient Response
- ② Steady state Response
- ③ Zero Input Response
- ④ Zero state Response.

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